higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

COMMUNICATION-ELECTRONICS N4

28 MARCH 2019

This marking guideline consists of 8 pages.

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QUESTION 1

1.1 False
1.2 True
1.3 False
1.4 False
1.5 True
1.6 True
1.7 False
1.8 True
1.9 False
1.10 False

(10 × 1) **[10]**

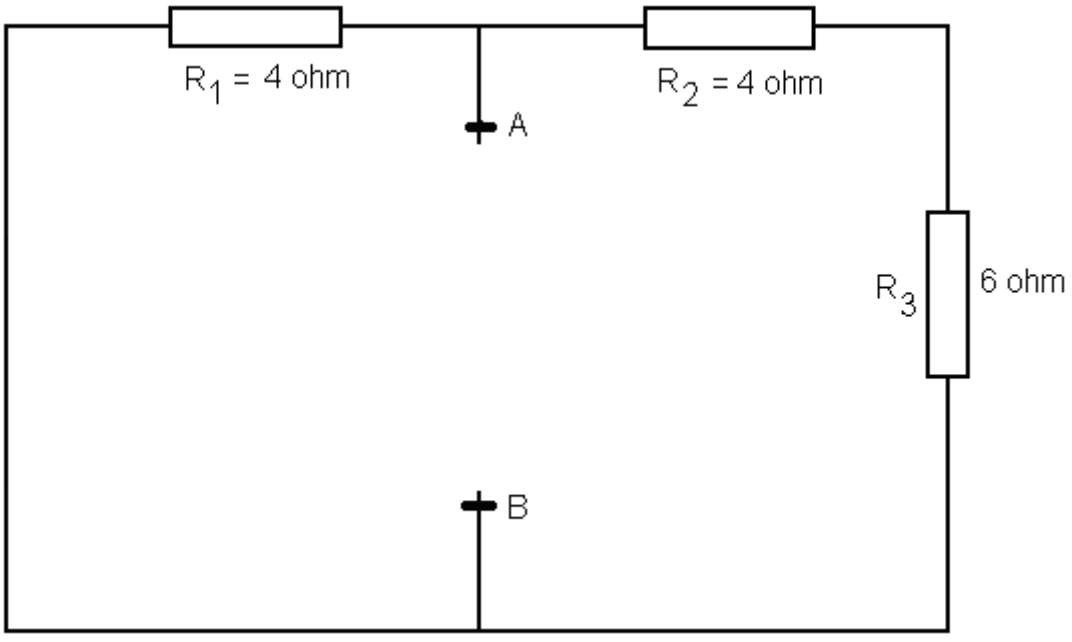
QUESTION 2

2.1 2.1.1 The algebraic sum of the currents flowing towards a point is equal to the sum of the currents flowing away from that point.

2.1.2 Any two terminal networks of impedances and generators may be replaced by a single equivalent voltage generator and series impedance.

(2 × 2) (4)

2.2 Remove load and short-circuit battery

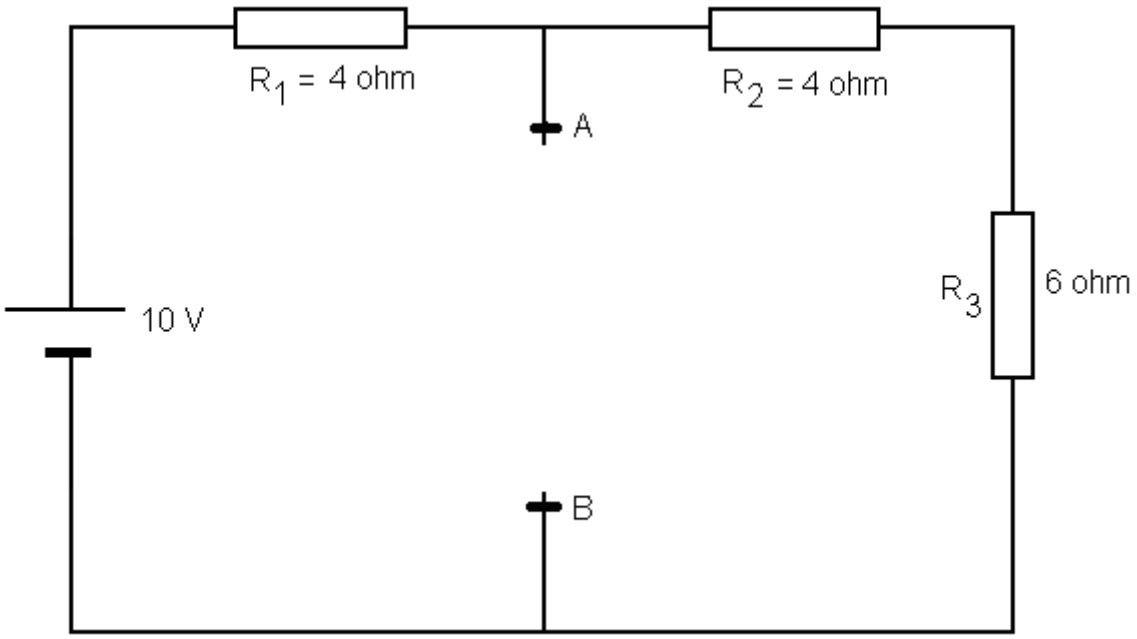


$R_{TH} = R_{AB} = R_1 \times R_s$
$R_1 + R_s$

$$= \frac{4 \times 10}{4 + 10}$$

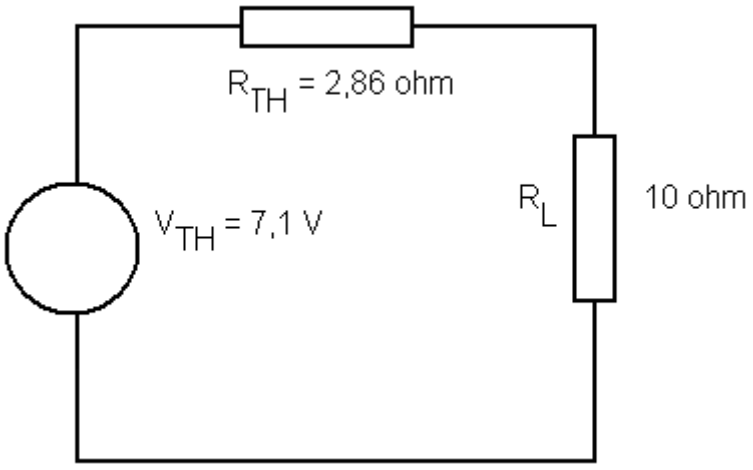
$= 2{,}86\ \Omega$

Replace battery and leave AB open



$R_T = R_1 + R_2 + R_3 = 4 + 4 + 6 = 14\ \Omega$
$I_T = V/R_T = 10/14 = 0,71\ A$
$V_{AB} = V_{TH} = I_T \times R_s = 0,71 \times 10 = 7,1\ V$

Thévenin's equivalent circuit



(6)
[10]

QUESTION 3

3.1 3.1.1 $Z_D = L/CR$
$= 15 \times 10^{-3}/10 \times 200 \times 10^{-9}$
$= 6\ k\Omega$

3.1.2 $Fr = 1/(2\pi \sqrt{LC})$
$= 1/(2\pi \sqrt{15 \times 10^{-3} \times 200 \times 10^{-9}})$
$= 2{,}91\ kHz$

3.1.3 $B\text{-}W = R/(2\pi L)$
$= 10/(2\pi \times 15 \times 10^{-3})$
$= 102{,}1\ Hz$

3.1.4 $Q\text{-factor} = R^{-1}(\sqrt{L/C})$
$= 10^{-1}(\sqrt{12 \times 10^{-3}/220 \times 10^{-9}})$
$= 27{,}4$

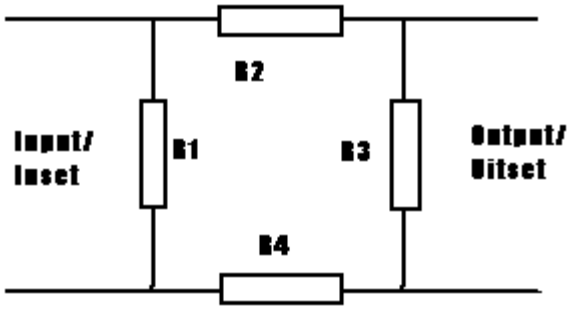
(4 × 2) (8)

3.2 $Q\text{-factor} = 2\pi f L/R$
$= 2\pi \times 2{,}91 \times 10^{3} \times 15 \times 10^{-3}/10$
$= 27{,}43$ (2)

[10]

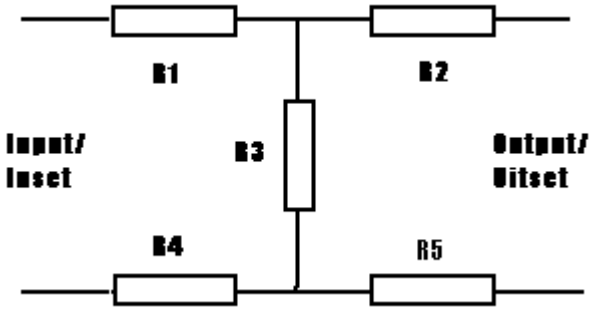
QUESTION 4

4.1 4.1.1



O-type attenuator

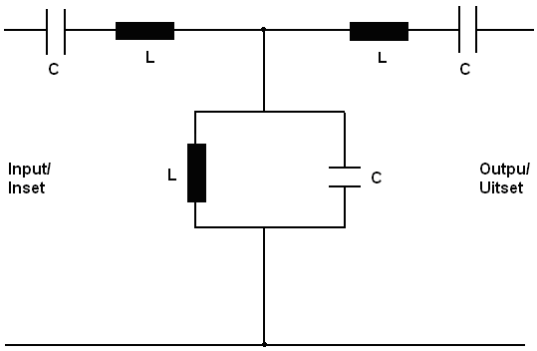
4.1.2



H-type attenuator

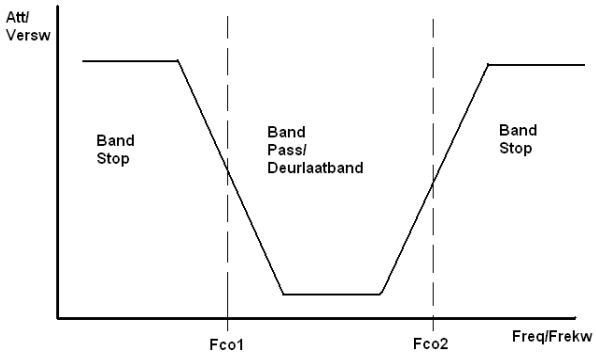
(2 × 2) (4)

4.2



Band-pass filter (3)

4.3

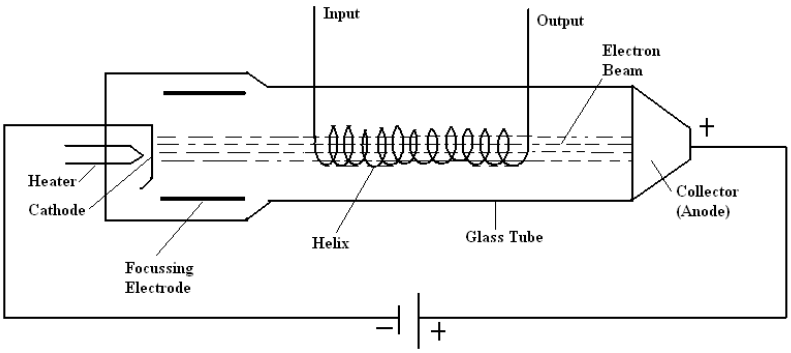


Attenuation versus frequency for a band-pass filter (3)

[10]

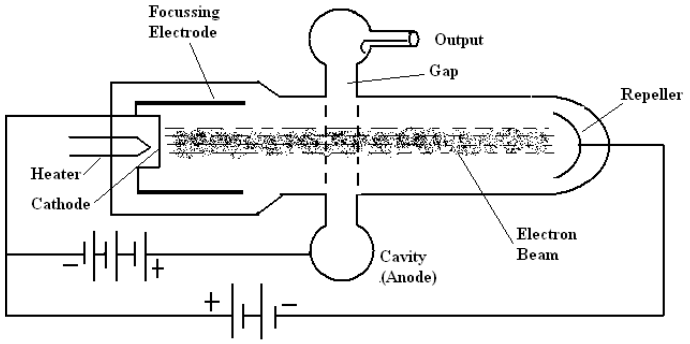
QUESTION 5

5.1



Travelling wave tube

5.2

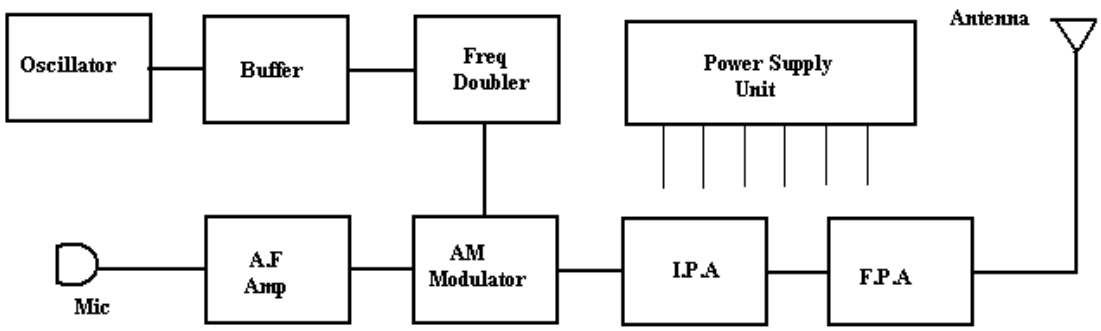


Reflex oscillator

(5 × 2) **[10]**

QUESTION 6

6.1

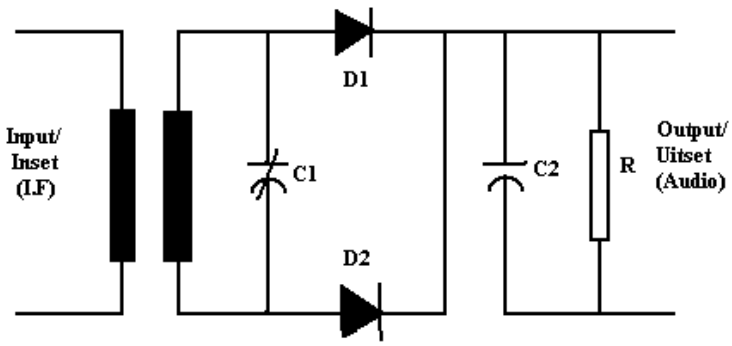


AM-transmitter (5)

6.2

- Oscillator: supplies RF signal (carrier) of constant amplitude and frequency
- Buffer: amplifies a small amount only and stabilises the oscillator output
- Frequency doubler: doubles frequency (tuned to second harmonic)
- Microphone: sound waves are converted to electrical currents
- Speech amplifier: amplifies the signal from the microphone
- Modulator: Input: audio signal + carrier and Output: modulated carrier wave
- IPA: amplifies the AM signal
- FPA: amplifies the modulated carrier to have sufficient energy to drive antenna circuits
- Antenna: converts electric currents into electromagnetic waves which are radiated into free space
- Power-supply unit: supplies regulated DC voltage to all stages (10)

6.3

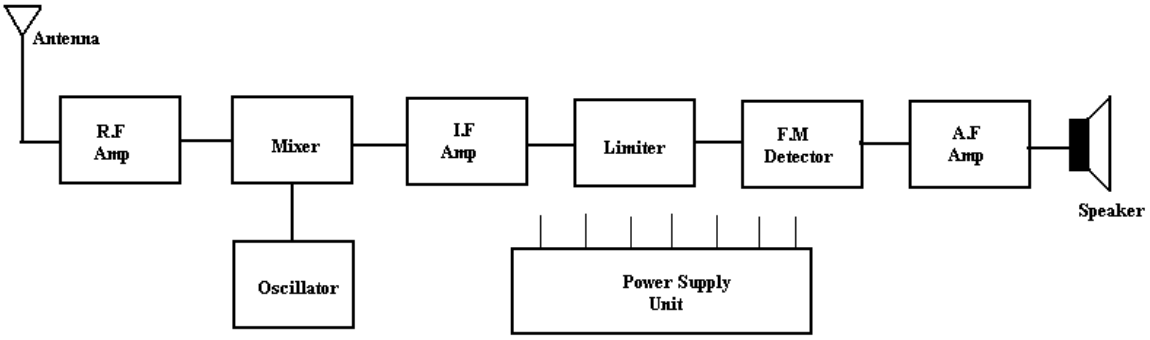


AM-full-wave detector (5)

[20]

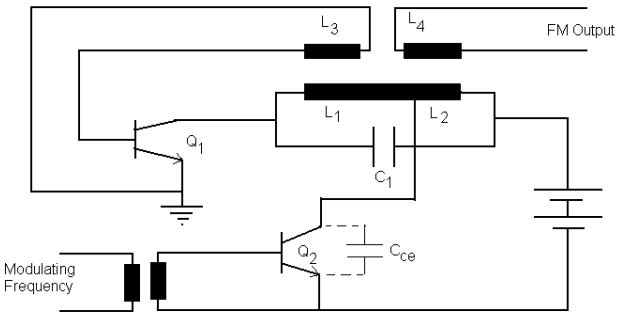
QUESTION 7

7.1



FM-superheterodyne receiver (5)

7.2



FM-modulator (5)

7.3 A process whereby the frequency of the carrier wave is made to vary in accordance with the instantaneous value of the modulating signal (2)

7.4 The degree of modulation in an FM-system (2)

7.5 To convert audio intelligence into a reactance (1)
[15]

QUESTION 8

8.1 $\lambda = v/f$

$= \frac{3 \times 10^8}{60 \times 10^6}$

$= 5$ m

Length of the antenna is 5/2 = 2,5 m (2)

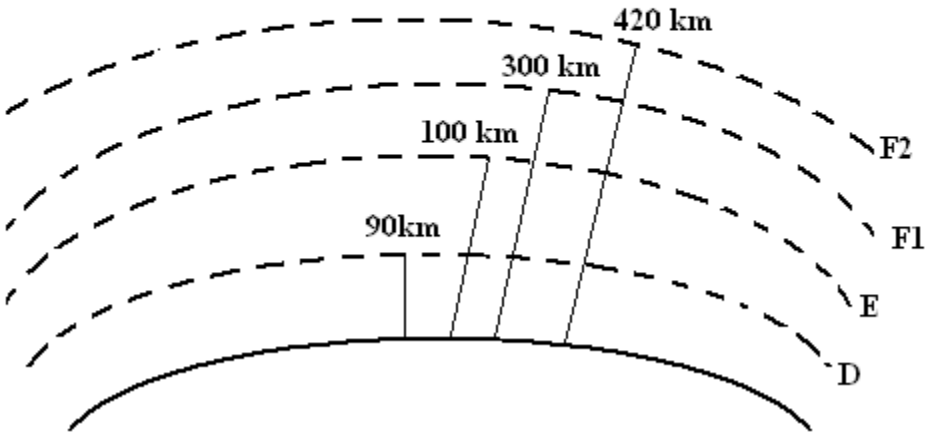
8.2 Electric field is vertically parallel to the antenna (1)

8.3 The ionosphere is the region in the upper atmosphere in which radio waves are bent back to the earth's surface. It is an area where the air pressure is so low that free electrons and ions exist for long periods without getting close enough to recombine into a neutral atom. (2)

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8.4 Day: F region spilts up into F_1 and F_2

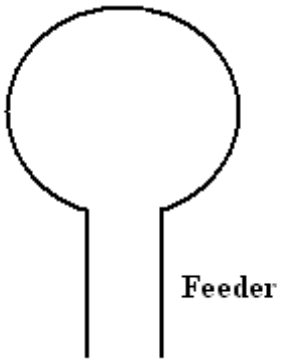
Night: F_1 layer disappears causing F_2 layer to drop



Ionosphere layers during the day as well as at night (4)

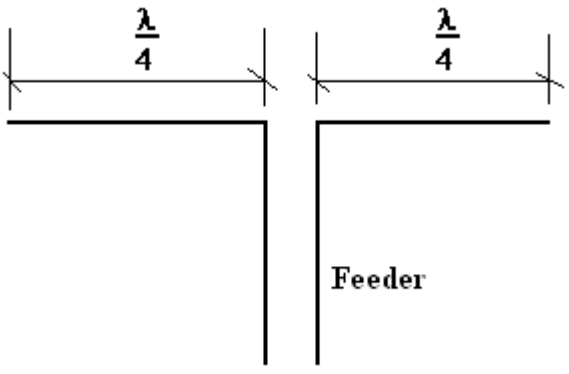
8.5 It is the angle between the norm and the radiated sky-wave for a given M.U.F. where the wave is first refracted back without penetration (2)

8.6 8.6.1



Loop antenna

8.6.2



Standard half-wave dipole antenna

(2 × 2) (4)

[15]

TOTAL: 100

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